

PRAKARSH AWASTHI

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EDUCATION

SRM Institute of Science and Technology	2024 - 2028	B.Tech, Computer Science	CGPA: 9.77
Narayana e-Techno School	2024	Class XII	77.2
St. Xavier's School	2022	Class X	89.4

EXPERIENCE

- **Research Project Intern - Indian Institute of Technology (ISM), Dhanbad** May 2026 – Present
 - Developing an AI-driven predictive maintenance framework featuring a tyre health inspection system based on YOLOv8, achieving over **97% classification accuracy** and **99% detection mAP** on a custom dataset of 10,000+ tyre images.
 - Engineered a multi-stage inspection pipeline comprising tyre localisation, damage classification, severity estimation, and automated health score generation for real-time condition monitoring.
 - Building a Digital Twin ecosystem for vehicle asset analytics and collaborating on the integration of hydraulic system diagnostics, motor Remaining Useful Life (RUL) prediction, and additional industrial health-monitoring modules for predictive maintenance applications.
- **Summer Research Intern - National Institute of Technology, Kurukshetra** May 2026 – Present
 - Researching Source Camera Model Identification (SCMI), leveraging forensic camera fingerprints and data-driven representations for robust camera model attribution.
 - Proposed and developing **WFGNet (Wavelet Fingerprint Graph Network)**, integrating Haar wavelet-guided patch ranking, ConvNeXt feature encoding, graph construction, and Graph Transformer refinement to model inter-patch forensic relationships.
 - Achieved **87.48% patch-level** and **88.27% image-level accuracy** on the Forchheim forensic dataset, with ongoing experimentation focused on graph-based representation learning and transformer-enhanced fingerprint analysis.
- **ML Research Intern - Snet Solutions, Puducherry** Oct 2025 – Dec 2025
 - Designed an end-to-end micro-expression-based deception detection framework incorporating onset-apex frame localisation, RAFT optical flow estimation, Reynolds motion-map generation, and deep feature learning for behavioural analysis.
 - Developed and benchmarked Bilinear Deep Convolutional Neural Networks (BDCNN), attention-augmented architectures (CBAM), and semantic Action Unit (Py-Feat) fusion strategies, achieving up to **95.8% deception detection accuracy**.
 - Performed extensive ablation studies on loss functions (Cross Entropy, Weighted Cross Entropy, ISDA-W), attention mechanisms, and feature representations, demonstrating that class-aware optimisation significantly outperformed architectural complexity under limited-data regimes.
- **Web Interface and Graphics Intern - Dexter Ventures Pvt Ltd, Delhi** June 2025 – July 2025
 - Designed and developed the complete Shopify-based e-commerce platform for Guitar World Store, implementing custom Liquid templates, responsive UI components, product catalog workflows, and optimized storefront architecture.
 - Engineered dynamic front-end features using HTML, CSS, JavaScript, and Shopify Liquid, enhancing navigation, product discovery, mobile responsiveness, and overall user experience.
 - Collaborated with stakeholders to translate business requirements into scalable web solutions, optimising performance, visual consistency, and customer engagement across the platform.

PROJECTS

- **AgniShakti – AI-Powered Fire Detection and Emergency Response Platform**
 - Developed a real-time fire and smoke detection system using custom-trained YOLOv11x models, achieving **mAP ≥ 0.85** on live surveillance feeds.

- Engineered a multi-camera monitoring and incident localization framework with automated alert generation, including timestamps, snapshots, confidence scores, and location metadata.
- Built a role-based emergency response dashboard for live incident tracking, society-wide monitoring, and rapid coordination between property owners and fire stations; awarded **2nd Place at Code Kalari National Hackathon (50,000)**.
- **Ascleon – Clinically Interpretable ECG Arrhythmia Detection System**
 - Developed a six-stage ECG intelligence pipeline on the PTB-XL dataset (21,837 recordings), incorporating signal denoising, multi-domain feature engineering, and automated arrhythmia classification for clinical decision support.
 - Engineered statistical, morphological, frequency-domain, and HRV-based cardiac biomarkers, and optimized an XG-Boost classifier using Optuna-driven hyperparameter tuning, probability calibration, and threshold optimization.
 - Achieved **91.54% accuracy**, **ROC-AUC of 0.9536**, and **F1-score of 0.784**, outperforming Logistic Regression, SVM, Random Forest, Ensemble Stacking, and a 1D ResNet baseline while enabling sub-millisecond CPU inference through a FastAPI deployment interface.
- **VoxSentinel – Real-Time AI Voice Fraud Detection Platform**
 - Developed a low-latency voice fraud detection system that continuously analyses live audio streams and detects imperceptible AI-generated artifacts during active calls.
 - Engineered a multi-model inference pipeline integrating SSL-AAIST, Whisper Encoder, AST Deepfake Detection, and forensic DSP analysis with weighted decision fusion for robust risk assessment.
 - Built a real-time forensic dashboard featuring spectrogram visualization, anomaly heatmaps, dynamic risk scoring, audit logging, and automated PDF/JSON investigation report generation.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL
Databases: MongoDB, MySQL, Firebase
Frameworks: PyTorch, TensorFlow, OpenCV
Tools: Git, GitHub, Linux, Kaggle

Web: React.js, Node.js, Express.js
AI/ML: Deep Learning, CV, SCMI
Research: Graph Learning, Digital Twins
Concepts: DSA, OOP, DBMS, OS, CN

POSITIONS OF RESPONSIBILITY

- **Committee Head – Directorate of Student Affairs, SRMIST:** Emceed events at **Milan**, one of India's largest national-level cultural festivals, performing before audiences of nearly **50,000 attendees**, while coordinating large-scale student initiatives and programs.
- **Committee Head – Aaruush, SRMIST:** Organized and managed events at **Aaruush 2026**, one of India's largest national-level techno-management festivals, overseeing event operations and participant engagement.
- **Music Club Secretary – Directorate of Hostels, SRMIST:** Led the college band to the **Finals of Battle of Bands at IIIT Kancheepuram** and performed across multiple university and inter-collegiate stages while organizing cultural events.
- **Project Team Lead – E-Yantra Lab, IIT Bombay:** Leading a multidisciplinary student team to solve robotics and automation problem statements, coordinating project execution under faculty mentorship.

CERTIFICATIONS & ACHIEVEMENTS

- **2nd Place** – Code Kalari National Hackathon, IIIT Kottayam (**50,000 Prize**)
- **6th Place** – IIT Guwahati National Hackathon
- **Finalist** – Global Digital Health Summit (GDHS) International Hackathon, Mumbai
- Top Achiever for Academic Year 2024–2025, SRMIST
- Introduction to Computer Vision and Image Processing – IBM
- Programming in Java – IIT Kharagpur, NPTEL
- Fundamentals of Object Oriented Programming – IIT Roorkee, NPTEL
- Reasoning Ability Mastery Logical Reasoning — Udemy
- Full Stack Development – GeeksforGeeks
- Best Musician Award – SRM Institute of Science and Technology